Stuck in the Matrix: Probing Spatial Reasoning in Large Language Models

Maggie Bai, Ava Kim Cohen, Eleanor Koss, Charlie Lichtenbaum*†

October 24, 2025

Abstract

This paper explores the spatial reasoning capability of large language models (LLMs) over textual input through a suite of five tasks aimed at probing their spatial understanding and computational abilities. The models were tested on both fundamental spatial reasoning and multi-step problem-solving within structured grid-based environments using tasks such as quadrant identification, geometric transformations, distance evaluation, word searches, and tile sliding. Each task was scaled in complexity through increasing grid dimensions, requiring models to extend beyond simple pattern recognition into abstract spatial reasoning. Our results reveal that while LLMs demonstrate moderate success in all tasks with small complexity and size, performance drops off rapidly as scale increases, with an average loss in accuracy of 42.7%, and reaching as high as 84%. Every test that began with over 50% accuracy showed a loss of at least 48%, illustrating the consistent nature of the deterioration. Furthermore, their struggles with scaling complexity hint at a lack of robust spatial representations in their underlying architectures. This paper underscores the gap between linguistic and spatial reasoning in LLMs, offering insights into their current limitations, and laying the groundwork for future integrative benchmarks at the intersection of language and geometry.

1 Introduction

Large language models (LLMs) have achieved remarkable success across a wide array of language-related tasks, including text generation, summarization, and question answering [3] [11], and many benchmarks, such as TruthfulQA [8], have been developed to evaluate this performance. Their ability to handle non-linguistic reasoning tasks, particularly those involving geometry and spatial relationships, has received significant attention from research into Visual Language Models (VLMs), both on training methods [4] and benchmarks [12]. Yet less work has been done on the intersection of the two, namely an LLM's spatial reasoning ability over text-based inputs.

Moreover, we observed in attempting to have LLMs play simple games like 2048 or Connect 4 through prompt-based board inputs that while models understood the strategy and gameplay, they often still made incorrect or invalid moves due to a failure to comprehend the board state. This manifested both in the initial reading of the board and in the application of principles such as gravity in Connect 4, emphasizing the limitations of text-based spatial reasoning in LLMs. As such, while the issues could have been solved by switching to a VLM, we decided to investigate the extent of LLMs' spatial reasoning abilities in text-based domains. These capabilities may improve LLM usage and reliability in layout-heavy fields such as finance, where interpreting tabular data correctly is necessary.

In this study, we designed a series of simple, isolated tests to measure these capabilities, with each task progressively increasing in complexity. These tasks include:

*Correspondence to: Charlie Lichtenbaum (clichtenbaum@gmail.com)
†Alphabetical ordering, all authors contributed equally

1. Quadrant: determining which quadrant of a Cartesian grid contains a specific symbol.
2. Transformation: reflecting a symbol across axes and producing the correct spatial configuration.
3. Distance: identifying the closest and farthest symbols relative to a target point on a grid.
4. Word Search: locating a defined word within a grid
5. Slide: sliding an X in a cardinal direction on a grid until it hits the edge or a wall

We evaluated four LLMs: GPT-4o, GPT-4.1, Claude 3.7 Sonnet with no thinking tokens (referred to as No Thinking), and Claude 3.7 with 16,000 thinking tokens (referred to as Medium Thinking). Each task was applied to progressively larger grid sizes, enabling detailed analysis of how well models generalize simple spatial reasoning to larger structures and more intricate configurations.

2 Related Works

This paper was inspired by work done with LLMs on text-based games, where experiments showed that models were unable to form and maintain the game worlds [14]. In addition, in Topaskal et al. [13], models were presented with boards for Tic-Tac-Toe, Gomoku, and Connect 4 in image and text-based formats. Their results showed a higher rate of invalid moves on more complex grid states in both the image and "illustration," or ASCII grid, formats, implying that the LLMs failed to comprehend the board properly. As such, this study sought to identify the complexity at which models begin to deteriorate as well as the types of problems the models encounter.

LLMs are also frequently benchmarked on their geometric abilities. Datasets of math problems, such as MATH [7] or GPSM4K [1], have been used to test the multi-step reasoning and geometry capabilities of various LLMs. These studies have all found the space for significant improvement in LLM spatial processing and reasoning.

However, these studies differ from ours in two significant ways. Firstly, this study was not designed to push the limits of the mathematical capabilities of the LLMs, unlike the datasets of Olympiad or graduate-level novel geometry problems. Rather, it tests basic reasoning and understanding of relative placements, with only simple arithmetic involved. And secondly, the models in our study receive their inputs in the form of ASCII grids, as opposed to images or textual descriptions of layouts. As such, this study serves better as an inquiry into tokenization and interpretation by the models.

Another benchmark in LLM spatial reasoning is the competition ARC-AGI [5], which has spurred numerous developments in LLM spatial reasoning capabilities over the six years since its release. The recently released ARC-AGI-2 [6] will likely continue to push researchers towards further developments. This paper, however, was not largely focused on developing novel strategies for enhancing reasoning capabilities, but rather benchmarking pre-existing ones.

Finally, a similar problem to spatial reasoning is that of tabular reasoning, or the interpretation of mixed text and layout documents. This kind of reasoning is common to fields such as finance or business, where contracts and invoices often contain tabular data. Researchers have developed specific tabular models, such as DocLLM [15], to fit these tasks, and have shown significant improvement compared to standard LLMs. This is a similar problem to that of spatial reasoning, as understanding the structure and relative positioning of elements in a layout is imperative to accuracy, and a parsing mistake can cost a company thousands of dollars. As such, improving the spatial reasoning capabilities and reliability of LLMs could increase their use cases significantly.

3 Methodology

To evaluate the spatial reasoning capabilities of the models, we designed five distinct tests that each targeted a specific aspect of spatial understanding. All tasks used square grids composed of simple symbols (for example, ‘·’ or letters) and were progressively scaled to increase complexity. All grids were presented with spaces as delimeters between indices. An example is shown below; all grids tokenized in the same manner, just with alternate characters instead of “."s.

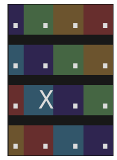

Figure 1: Example Tokenization

3.1 Quadrant

In this task, a single ‘X’ was placed randomly on a grid, and the models were asked to determine the quadrant of the grid in which the ‘X’ was located. The quadrants were defined in alignment with the Cartesian coordinate system and the models replied in a “Lower/Upper Left/Right" format. An example grid is shown below:

$$\begin{array}{cccc} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & X & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array}$$

Figure 2: In this example, the ‘X’ is in quadrant III (lower left quadrant).

3.2 Transformation

This task tested the ability of the models to perform geometric transformations. A single “X" was randomly placed on a square grid, and the models were asked to reflect the “X" over a centerline (not given), either horizontally or vertically, and return a coordinate pair in (x, y) format. Example reflections are shown below:

$$\begin{array}{cccc} \cdot & \cdot & \cdot & \cdot \\ X & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array} \longrightarrow \begin{array}{cccc} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ X & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array}$$

(a) Reflection vertically.

$$\begin{array}{cccc} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & X & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array} \longrightarrow \begin{array}{cccc} \cdot & \cdot & \cdot & \cdot \\ \cdot & X & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array}$$

(b) Reflection horizontally.

3.3 Distance

In this task, letters A-G and an X were placed on a square grid. The models were asked to identify which letter was closest to and farthest from the X. An example grid is shown below: